

Highlights

Structurally prominent linguistic items hold a privileged state in working memory: Time-course evidence for rapid access

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- Using a published visual world dataset, we compare reactivation of structurally prominent targets vs. distractors for a Turkish anaphor.
- Target referents are accessed more rapidly and more robustly.
- Distractors' reactivation is weaker and temporally delayed.
- Structural information organizes linguistic items in working memory.
- Targets are held in a privileged, focal state, available for immediate access.
- Distractors lack this privilege and rely on slower retrieval for reactivation.

Structurally prominent linguistic items hold a privileged state in working memory: Time-course evidence for rapid access

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Abstract

Real-time language processing relies on a capacity-limited working memory system to encode and maintain (extra-)linguistic information. Prior research shows that structural information is rapidly used to distinguish targets from distractors in memory access, but the exact mechanisms underpinning this remain unclear. Here we consider two hypotheses. One is that structurally prominent, target representations are actively maintained in a privileged, focal attentional state that allows rapid, direct access. Another possibility is that structural prominence serves as a retrieval cue to distinguish target representations stored in a passive, non-focal state. These hypotheses can be distinguished by determining the relative time course of memory access to targets vs. distractors. To do this, we conducted a secondary analysis of a published dataset that measured the looking behavior during the online resolution of the structurally constrained Turkish local reciprocal *birbirleri* in a visual world paradigm. Bayesian analyses on gaze proportions and new fixations show evidence for reliable and faster access to targets relative to distractors. We interpret this as evidence for a functional divide within linguistic memory that is shaped by hierarchical syntactic structure, whereby structurally prominent target items are held in a privileged focal state, while distractors that lack this privileged status are accessed via a slower retrieval mechanism.

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1. Introduction

Real-time language processing requires encoding novel representations of the linguistic input in working memory, followed by the maintenance and retrieval of these representations over time. This process is necessary for the formation of linguistic dependencies that span multiple words or phrases. An example is the dependency between a local anaphor, such as *each other*, and its antecedent, *lawyers*, as in ‘*The lawyers of the suspects secretly liked each other*’. In the online interpretation of such local anaphors, constraints defined on the basis of configurational syntactic structure are rapidly used in memory access to distinguish structurally available *target* antecedents (e.g., *lawyers*) from unavailable *distractors* (e.g., *suspects*) (Dillon et al., 2013; Kush and Phillips, 2014; Parker and Phillips, 2017; Parker, 2019a; Sturt, 2003; Xiang et al., 2009) (for a recent review, see Dillon and Keshev, 2024). However, it remains an open question how such syntactic structure is leveraged within working memory to do this.

Well-established models of working memory posit a distinction between a capacity-limited, active, *focal* state, in which representations remain continuously available for immediate access, and a passive, *non-focal* state that requires slower, explicit retrieval (Broadbent, 1958; Cowan, 2010; McElree and Doshier, 1989; McElree, 1996, 1998, 2001, 2006; Oberauer, 2002, 2009, 2019; Wickelgren et al., 1980). Evidence for the focal/non-focal divide comes primarily from studies investigating the temporal dynamics of memory access. In various list-memorization tasks, this division manifests as a distinct speed advantage for the most recently processed item (McElree, 1996, 1998, 2006; McElree and Doshier, 1989, 1993) or the specific item cued for recall (McElree, 2001). These studies demonstrate that information can be strategically brought into a focal state and actively maintained there while processing intervening material. This functional division is further corroborated by neuroimaging evidence, which identifies divergent neural signatures associated with the processing of focal and non-focal information (Öztekin et al., 2008, 2010).

A core property of the focal state is its limited capacity, which many accounts restrict to a single task-relevant representation (e.g., Broadbent,

1958; McElree and Doshier, 1989; McElree, 2001, 2006; Oberauer, 2002, 2009); cf. Cowan, 2010). Crucially, what constitutes a single representation does not necessarily correspond to an individual atomic ‘item’. Instead, it can be better defined as a single functional context integrated via properties such as shared time, space or other stimulus characteristics (Jonides et al., 2008; McElree, 1998). Supporting evidence for this comes from McElree (1998), who demonstrated that when participants were encouraged to chunk a nine-word list into three semantic triplets, the retrieval speed advantage typically reserved for the final item extended to all three words within the most recent semantic category.

While the focal/non-focal divide determines various memory dynamics such as speed and accuracy, mapping this architecture onto real-time language processing is not straightforward (Lewis and Vasishth, 2005; McElree, 2000; McElree et al., 2003; Wagers and McElree, 2013). Unlike recalling disconnected word lists, sentence comprehension requires building novel structural representations of the linguistic input. It is unclear exactly how linguistic input is chunked into discrete memory representations, and which of those chunks constitutes the active focus of attention (Wagers and McElree, 2013).

A growing body of experimental work gives some clues. Certain types of linguistic information, such as syntactic category, semantic content or grammatical number, appears to be maintained in a focal state during on-line processing (Franck and Wagers, 2020; Johns et al., 2015; McElree, 2000; McElree et al., 2003; Ness and Meltzer-Asscher, 2019; Wagers et al., 2009; Wagers and Phillips, 2014; Wagers and McElree, 2022). For example, speed-accuracy trade-off (SAT) studies have investigated the speed and accuracy of accessing a subject noun in subject-verb dependencies (Johns et al., 2015; McElree et al., 2003; Wagers et al., 2009). These studies find that the subject noun is accessed significantly faster when it is adjacent to its verb or separated by a prepositional phrase as in ‘*The editor (of the book) laughed*’, compared to when it is separated by a relative clause as in ‘*The editor that the book amused laughed*’. This suggests that features of the subject can be maintained in a privileged state, but perhaps only within the immediate clausal domain. Extending this work, Franck and Wagers (2020) used jabberwocky stimuli to show that abstract syntactic features relevant for grammatical subjecthood can remain in focal attention in the absence of semantic information. Wagers and Phillips (2014) provided self-paced reading evidence that the broad syntactic category of a wh-filler (e.g., NP or PP), but not semantic or verb

subcategorization information, can be kept in an active, focal state in processing filler-gap dependencies over long distances. More recently, Wagers and McElree (2022) showed in an SAT task that individual morphosyntactic features of a determiner, such as plural number, can be maintained in the focus of attention during the processing of a determiner-noun dependency as in ‘*those clever, risk-taking burglars*’.

Drawing on these findings, we hypothesize that the advantage of structurally defined target representations in initial memory access may be due to their maintenance in a privileged, focal state in memory. Under this hypothesis, hierarchical syntactic information guides the organization of linguistic representations in memory during encoding, such that structurally prominent linguistic items are selectively maintained in a privileged, focal state. This would enable *faster*, direct access to these items relative to structurally less prominent, hierarchically embedded items, such as the target *lawyers* vs. the distractor *suspects* in ‘*The lawyers of the suspects secretly liked each other*’ (repeated from above).

This hypothesis stands in contrast to an alternative hypothesis that has guided much research: namely, that hierarchical syntactic information is instead deployed via retrieval cues in a structurally guided retrieval process. Under this hypothesis, all linguistic representations that exceed the span of focal attention, regardless of their structural prominence, are stored uniformly in a passive, non-focal state and are accessed via the same content-based associative retrieval mechanism (Lewis and Vasishth, 2005; Lewis et al., 2006; Martin and McElree, 2008, 2009, 2011; McElree, 2000; McElree et al., 2003; Van Dyke and Lewis, 2003; Van Dyke and McElree, 2006, 2011). Upon encountering a retrieval probe, a parallel search is initiated to check a set of retrieval cues, including both structural and featural (number, gender etc.) ones, against all passive representations (Dillon et al., 2013, 2014; Kush and Phillips, 2014; Kush et al., 2015; Parker and Phillips, 2017; Parker, 2019a; Wagers, 2008; Schoknecht et al., 2025; for a detailed discussion on structural cues, see Kush, 2013). Because all items are evaluated in parallel against the full cue set, this account predicts an initial, simultaneous reactivation of any referent that has a (partial) match, including distractors in structurally unavailable positions. While distractors will ultimately reach distinctly lower activation levels relative to targets due to their mismatch with structural cues, on this view this distinction should only emerge gradually as the retrieval process unfolds.

To evaluate these two hypotheses, we conduct a secondary analysis of

an earlier, published visual world eye-tracking dataset on the real-time processing of the structurally constrained Turkish local anaphor *birbirleri* ‘each other’ (Bakay et al., 2026). The experimental stimuli contrast target antecedents in structurally prominent positions—subjects and indirect objects—against distractors embedded in postpositional phrases. In these stimuli, structural prominence refers to those items that hierarchically dominate (e.g. c-command) the anaphor, or are otherwise co-arguments of the same verbal domain. All other items in a sentence were deemed not structurally prominent (see the next section for more information). Other sources of information, such as (plural) number, morphological case, linear position and syntactic clause membership were held constant across targets and distractors.

Previous analyses of this dataset show that, within the anaphor window, structurally defined target subjects and indirect objects are reactivated to a greater degree than structurally unavailable distractors, providing evidence that syntactic structure rapidly guides memory access (Bakay et al., 2026). However, those initial analyses examined eye-movement behavior aggregated over the entire anaphor window, and do not reveal exactly when they get reactivated. Here, we analyze the time-course of reactivation for the critical target and distractor referents starting from the onset of the anaphor to provide some evidence about their representational states in memory.

We hypothesize that if structural prominence determines which memory representations are maintained in a focal versus non-focal state, hierarchically prominent linguistic items such as the subject and indirect object of a clause will be held in a privileged, focal state while that clause is being processed, rendering them directly accessible on demand. In contrast, structurally less prominent, hierarchically more embedded items such as distractors will be maintained outside of focal attention, requiring slower retrieval to be accessed. This is schematized for the Turkish stimuli examined in the current study in Figure 1.

To measure the time course with which target and distractors are reactivated, we use the visual world paradigm. This has been extensively used to track the time-course of reactivation for linguistic referents, grounded in the linking hypothesis that visual attention is closely time-locked and probabilistically related to the mental activation of linguistic representations (Allopenna et al., 1998; Cooper, 1974; Huettig et al., 2011; Kamide et al., 2003a,b; Magnuson, 2019; Tanenhaus et al., 1995). It follows from this link that continuous eye-movement behavior can provide a real-time index of

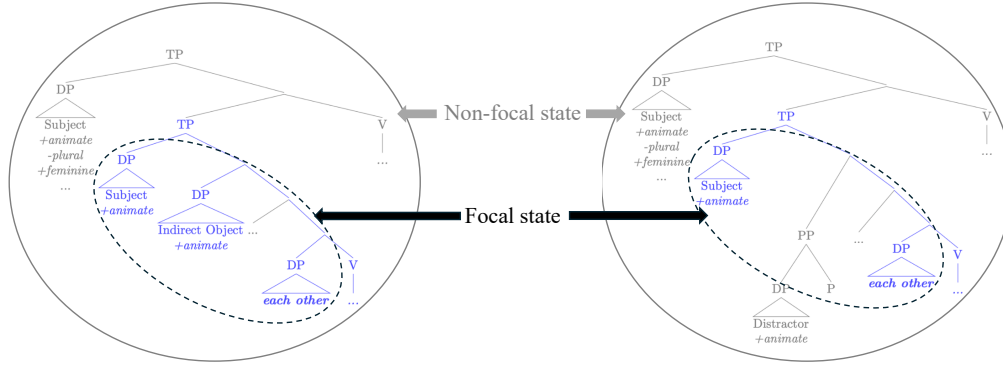


Figure 1: Schematization of the focal vs. non-focal memory states shaped by structural prominence at the point of processing *each other*, for the Turkish stimuli examined in the current study. The smaller, dashed circle includes the items selectively maintained in the focal state; and the larger, solid circle includes those outside of focal attention. The left panel illustrates a sentence with a hierarchically prominent subject and indirect object in focal attention; the right panel illustrates a sentence with a hierarchically less prominent, embedded distractor in the non-focal state.

memory access by recording shifts in visual attention towards linguistic referents over time. We leverage this property to measure whether hierarchically prominent targets are reactivated earlier than less prominent distractors.

We predict that if hierarchically prominent, plural targets are selectively maintained in a privileged, focal state as shematized in Figure 1, the eye-movement data should reflect a dichotomous temporal profile, with significantly earlier looks to the targets (i.e. subjects and indirect objects) relative to hierarchically less prominent distractors (i.e. NPs embedded in a PP). Conversely, if all representations are stored uniformly in a non-focal state and distinguished at retrieval via structurally informed cues, we should initially observe a comparable, simultaneous increase in early looks to both targets and distractors due to their (partially) matching features, followed by an accelerating bias towards the target as structurally informed retrieval cues differentially increase its reactivation.

To preview the results, we find that target referents in hierarchically prominent positions are reactivated significantly faster than distractors in hierarchically less prominent positions. This temporal asymmetry suggests that abstract structural information shapes the organization of representations in linguistic memory, such that syntactically prominent representations are maintained in a privileged focal memory state over the course of sentence

comprehension.

2. Materials and methods

We address our key question thorough a secondary analysis on a published, visual world dataset in Bakay et al. (2026). This dataset investigated the retrieval of hierarchically prominent target referents vs. distractors during the resolution of the Turkish local reciprocal *birbirleri* ‘each other’. Local anaphors provide an ideal testing ground for examining the role of structural information in working memory, as their interpretation is structurally constrained, i.e., they must be bound by antecedents in hierarchically prominent (e.g., c-commanding) positions within their local domain (e.g., clause) (Chomsky, 1981; see Akkuş, 2021; Kornfilt, 1997 for Turkish). We pool datasets from two experiments in the original study to maximize statistical power.

2.1. Participants

Data from 156 native speakers of Turkish (54 from Experiment 2 & 102 from Experiment 3 in the original study) were analyzed. Data from 4 participants were removed either due to low accuracy in the filler items or major track loss, leaving 152 participants for analysis. Each participant was paid 300 to 350 Turkish liras (~\$11 at the time of the study) for compensation.

2.2. Stimuli

The stimuli had two conditions, as exemplified in (1). In both conditions the reciprocal anaphor *birbirleri* ‘each other’ was the direct object of an embedded clause, whose subject was always a grammatically available, *target* antecedent for the anaphor. The conditions differed in the structural position of an additional noun phrase, *yönetmenler* ‘directors’. In the Indirect Object (IO) condition, this noun was in a hierarchically prominent position as the IO of the embedded clause; hence, a grammatically available antecedent for the anaphor. This led to an ambiguity between the subject and the IO as the antecedent of the anaphor. In the Distractor condition, the same noun was hierarchically subordinated in a prepositional phrase and hence served as a *distractor*,

The structural availability of the critical referents can be defined via their *c(onstituent)-command* relations with respect to the anaphor within the sen-

tence’s hierarchical configuration (Chomsky, 1981; Reinhart, 1976, 1983)¹. Formally, a node α c-commands a node β if β is α ’s sister or is contained within α ’s sister in the tree structure. Thus, while the embedded subject and IO are sisters to the verb phrase containing the anaphor and c-command it, the distractor is not a sister to that same verb phrase and therefore does not c-command the anaphor. The structural availability of all referents is marked via coindexation in (1).

- (1) Example item (with anaphor in bold and c-commanding target(s) in italics)
- a. C-COMMANDING IO
Ayşe_i [*kameraman-lar-in_j* *yönetmen-ler-e_k* açıkça
Ayşe cameraman-PL-GEN director-PL-DAT openly
birbirlerin-i_{*i/j/k} sor-duğ-un]-u anla-dı.
each.other-ACC ask-NMLZ-3SG.POSS-ACC notice-PST
‘Ayşe_i noticed that the cameramen_j openly asked the directors_k
about each other_{*i/j/k}.’
- b. NON-C-COMMANDING DISTRACTOR
Ayşe_i [*kameraman-lar-in_j* *yönetmen-ler-e_k* karşın açıkça
Ayşe cameraman-PL-GEN director-PL-DAT despite openly
birbirlerin-i_{*i/j/*k} sor-duğ-un]-u anla-dı.
each.other-ACC ask-NMLZ-3SG.POSS-ACC notice-PST
‘Ayşe_i noticed that the cameramen_j, despite the directors_k, openly
asked about each other_{*i/j/*k}.’

The embedded subject, IO and distractor were all plural, matching the anaphor in number. The IO and distractor were marked with the dative case $-(y)A$, and were in the same linear position, i.e., following the embedded subject. A manner or time adverb, such as *açıkça* ‘openly’, preceded the anaphor to allow more time for the processing of the critical referents and minimize potential bias in looks before the critical anaphor window.

¹While the structural constraints on the anaphor–antecedent dependencies examined here can also be formulated in terms of co-argumenthood (Pollard and Sag, 1992; Reinhart and Reuland, 1993), here we adopt the notion of c-command, as it plays a central role in syntactic theory and receives empirical support independent of co-argumenthood in the sentence processing literature (e.g., Kush et al., 2015; Moulton and Han, 2018).

The dataset for the present study was pooled across two source experiments, creating 54 unique item sets (comprising 30 and 24 sets from Experiments 2 and 3 in the original study, respectively). Due to additional conditions in the original designs, these items were distributed across multiple lists (5 lists in Experiment 2 and 3 lists in Experiment 3) using a Latin square design. Consequently, individual participants saw a subset of the total items that we analyze here: A single participant saw 12 out of 30 item sets in Experiment 2, and 16 out of 24 item sets in Experiment 3. This counterbalancing scheme ensured that all 54 item sets were represented in the pooled dataset.

Offline antecedent selection tasks that were conducted before the visual world studies confirmed that hierarchical structure constrains the interpretation of the anaphor: While c-commanding, target subjects and IOs can bind the anaphor, non-c-commanding distractors cannot (see the original study for details).

2.3. Procedure

Participants were tested individually in an eye-tracking lab located at a public university in Istanbul, Turkey. Eye movements were recorded at 1000 Hz using a desktop-mounted SR Research EyeLink 1000 Plus and a chin rest (SR Research, Ontario, Canada).

The study employed a goal-oriented visual world paradigm, as schematized in Figure 2. Each trial started with an auditory lead-in sentence, followed by a question that cued the participants to identify the referent of the anaphor in the upcoming experimental sentence. An example lead-in sentence and a question for the example item set in (1) are provided in (2). The question was presented before the experiment sentence to direct gaze towards the potential referents during the anaphor window as rapidly as possible. Following the question, a preview of the visual display was presented featuring four images: the subject, the IO/distractor, an unmentioned singular referent and an unmentioned plural referent, with each image being highlighted for 1.5 seconds. The preview was included to isolate gaze patterns driven by linguistic processing from those driven by visual search (Henderson and Ferreira, 2004). The trial ended once the participants clicked on the image that referred to the anaphor in the sentence.

- (2) a. Lead-in

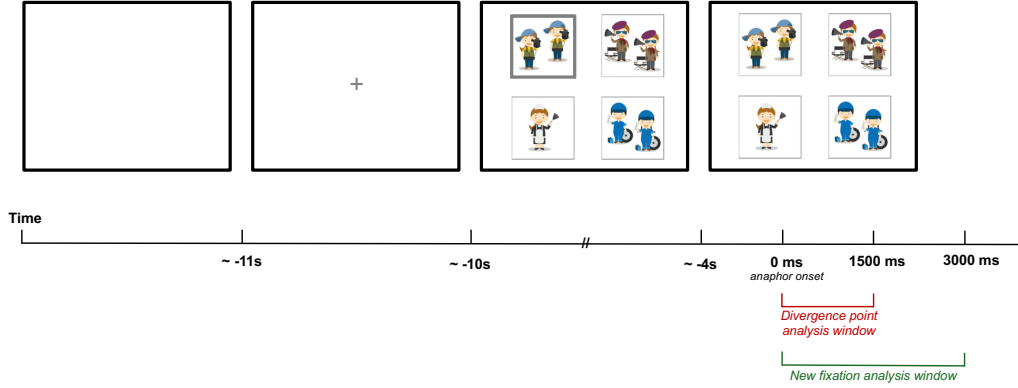


Figure 2: Schematic representation of the trial structure for the stimuli in (1). After the auditory lead-in sentence and question with a blank visual display, there was a drift correction and a 1000 ms fixation period. A 6-second visual preview followed, which highlighted the subject (top left), IO/distractor (top right), and unmentioned singular/plural referents (bottom), for 1.5 seconds each. During the auditory experimental sentence, the display remained on screen until a response selection. The timeline is relative to anaphor onset (0 ms), marking the time window for the divergence point (0-1500 ms) and new fixation (0-3000 ms) analyses. // indicates a non-proportional temporal scale.

- Ayşe meraklı soru-lar duy-muş.
 Ayşe curious question-PL hear-PST
 ‘Ayşe heard curious questions.’
- b. Question
- Peki Ayşe kim-in sorul-ul-duğ-un-u anla-dı?
 So Ayşe who-GEN ask-PASS-NMLZ-3SG.POSS-ACC notice-PST
 ‘So who did Ayşe notice was asked about?’

In addition to the experimental items, 72 fillers were included in which the anaphor was replaced with a name referent from the display.

2.4. Data analyses

The dataset was generated by preprocessing the output from EyeLink DataViewer using the `vwpre` package in R (Porretta et al., 2022). The eye-tracking data, which was originally sampled at 1000 Hz, were downsampled into temporal windows of 25 ms, 50 ms and 150 ms for the statistical analyses. Fixations that did not land on any of the four predefined interest areas were

categorized as ‘outside’ looks.

The time window of interest began at the onset of the *anaphor* and continued through the end of the trial. All analyses were time-locked to the anaphor onset (i.e., marked as time zero) for each item set. The onset of the critical window was shifted for 150 ms to account for the delay in monitoring eye movements in response to a stimulus (Rayner et al., 1983; Saslow, 1967). The average duration of the anaphor was 644.5 ms. While the duration of the anaphor remained constant across conditions in a given item set, it varied across sets due to differences in morphological case marking.

Analyzing visual world eye-tracking data poses several challenges due to the nature of the measurements: Fixations are binary, temporally autocorrelated and dependent across visual objects (Ito and Knoeferle, 2023; McMurray, 2023). To address these issues, we analyzed two complementary measures of eye movements, namely gaze proportions and new fixation probabilities. First, we analyzed gaze proportions to estimate the onset of divergence in the reactivation of critical referents. Second, we examined the time-course of new fixations launched towards these referents for a more precise estimate of the specific time points at which they are reactivated.

Analyses were conducted using Bayesian modeling with the **brms** package in R (Bürkner, 2017). For all statistical models, we report means of the posterior distributions over the parameter, their standard error and 95% credible intervals (CrIs; indicated by ‘[]’ in the text). The CrIs represent the range over which we can be (95%) certain that the values of the parameter lies given the statistical model and the data. Our statistical inferences are based on how much of the probability mass of the posterior distribution over each parameter lays above/below zero and the formal hypothesis testing for the presence or absence of an effect using Bayes factors (Lee and Wagenmakers, 2014; Schad et al., 2022). We calculated Bayes factors (e.g., BF_{10}) using the Savage-Dickey density ratio method (Dickey and Lientz, 1970; Dickey, 1971) by taking the inverse of the ratio of the posterior density to the prior density for a parameter of interest at zero. We interpret a posterior probability mass of 97.5% above/below zero or greater (i.e., $\Pr(\hat{\beta} > 0)$ or $\Pr(\hat{\beta} < 0) \geq 0.975$) as meaningful evidence for an effect. For Bayes Factors, we interpret BF_{10} values between 1 and 3 as anecdotal evidence, those between 3 and 10 as moderate evidence and those above 10 as strong evidence for the effect of interest; values below 1 denote evidence against the effect of interest (Lee and Wagenmakers, 2014). All models reported had $\hat{R}$ values of 1 for all parameters, and visual examinations of model fits indicated no issues with

convergence.

2.4.1. Divergence point analyses on gaze proportions

We conducted divergence point analyses on gaze proportions to identify when looks to different referents began to diverge. We employed a non-parametric bootstrap technique on gaze proportions over time following the procedures in Stone et al. (2020) and Minor (2026). While we adopted the parameter settings from Stone et al. (2020), we conducted the resampling at the participant level, following Minor (2026) (see below). This by-participant bootstrap preserves the structure of the original data by sampling all data points from a given participant intact, while allowing for generalization beyond the current sample². We also integrated Bayesian principles into the analyses following the approach in Stone et al. (2021).

We estimated divergence onsets by comparing looks towards the IO vs. distractor, either against the target subject or an unmentioned plural referent on the display. Comparisons against the subject measure when the two referents are distinguished from one another. While the divergence of the distractor from the subject reveals when hierarchical information is deployed to differentiate structurally available from unavailable referents, the divergence of the IO from the subject marks the resolution of the referential ambiguity between two grammatically available antecedents. Comparisons against the unmentioned referent indicate when the IO/distractor is reactivated relative to a baseline, plural referent that should not be reactivated by the anaphor in any condition.

The total critical time window for the analyses was set from the onset of the anaphor and extended for 1500 ms, which includes the processing of both the anaphor and the following embedded verb, and an additional 150 ms for saccade planning. This duration was initially motivated by a visual inspection of the gaze data, which suggested that the looks to critical referents likely diverge within this window. This was later confirmed in the analyses (see below), with divergence onsets reliably estimated within this window across all simulated datasets.

²We also conducted divergence point analyses using the resampling method with stratification by participant and time as in Stone et al. (2020). The results of these analyses were qualitatively similar to those reported here. We chose to report the results of the by-participant resampling, as it preserves the clustering of the data by trials, items and participants, and it provided more conservative estimates.

For the analyses we created a dataset with 50ms time bins³ in which participants’ fixations on the visual referents were coded binomially (fixated = 1; not fixated = 0) depending on whether or not they fell in the pre-defined region of interest at any point within the respective 50ms window. For each comparison, we generated 2,000 bootstrap datasets by resampling participants with replacement, stratified by Latin square list to preserve the original sample size per list. We then applied an uncorrected, one-sample *t*-test at each time point aggregating over items to reveal whether there were more looks towards one referent over the other. The divergence point was then defined as the first time point of 4 consecutive 50ms time bins (=200ms) in which the *t*-tests yielded significant results ($p < 0.05$). This was done for the IO and Distractor conditions separately in each comparison.

To compute posterior distributions for the divergence onset estimates, we approximated the likelihood function by fitting a normal distribution to the empirical distribution of onset estimates obtained via bootstrapping. For the prior distributions, we reasoned that divergence onsets should arise within the 1350 ms critical window spanning from the onset of the anaphor to the offset of the verb (after the 150 ms shifting for saccade planning). Our prior was thus a normal distribution centered within this critical window, with 95% of its probability mass falling between 0 and 1350 ms (i.e., $N(675, 344)$). The posterior distributions for each condition were then derived as the product of this prior and the normal approximation of the likelihood function (Smith, 2011). Because identical, independent priors were used for each condition, the resulting prior distribution on the between-condition difference was centered at zero, with symmetric probability mass over positive and negative differences (Weisstein, 2020).

We also conducted formal hypothesis testing using Bayes factors to quantify the degree of evidence for a between-condition difference. Because Bayes factors can be sensitive to the choice of prior distributions (e.g., Nicenboim et al., 2026; Schad et al., 2022), we performed a prior sensitivity analysis evaluating three different prior sets for the divergence onsets, as shown in

³We also ran divergence point analyses using 20 ms time bins, which yielded estimates that were delayed by up to 100 ms compared to those with the 50 ms time bins reported above. Based on simulated data, Fernandez et al. (2025) demonstrate that divergence onset estimates generated from 50 ms time bins are closer to the true divergence points than those generated from 20 ms bins. Given our own observations and those reported in Fernandez et al., we report the analyses with 50 ms time bins above.

Table 1. For each analysis, independent normal priors were assigned to the divergence onsets of the individual conditions, such that 95% of their probability mass fell between 0–1350, 0–1000 or 0–650 ms, corresponding to large, medium and small prior widths, respectively.

Prior width	Prior distribution	Assumed range (ms)
Large	Normal(675, 344)	[0, 1350]
Medium	Normal(500, 255)	[0, 1000]
Small	Normal(325, 166)	[0, 650]

Table 1: Prior specifications used for the Bayes factor sensitivity analyses on divergence points. Independent normal priors were defined for divergence onsets in individual conditions, with assumed ranges representing 95% of the prior probability mass on the ms scale.

2.4.2. Time-course analyses on new fixations

One drawback of traditional gaze proportion analyses is their reliance on gaze location, which may reflect the ‘cumulative history’ of eye movements rather than real-time eye-movement behavior. Eye fixations usually last 150 to 500 ms, with 200-250 being the average (Rayner, 1989, 2009). Consequently, a statistically significant effect in a given window may, at least in part, be a persistent artifact of an effect in preceding time windows where the experimental manipulation was active. To isolate effects on current eye movements from those that are due to fixations extending from previous time windows, we conducted additional analyses on the proportions of new eye fixations launched towards the critical referents within discrete time windows (also see Kingston et al., 2016).

In these analyses the dependent measure was the proportion of a new fixation launched towards the IO/distractor out of all new fixations towards the critical referents in the sentence, i.e., IO, distractor and subject. Trials without any new fixations towards these referents were excluded from the analyses. To restrict the analyses to *new* fixations only, we excluded fixations towards a referent if the relevant trial already started with looks to that referent at the beginning of the critical time window. While consecutive fixations on the same referent were counted as one using the time point of the very first fixation, non-consecutive fixations to a single referent were counted as distinct fixations. For example, if the fixation sequence in a given trial was Subject → Subject → IO → Subject, the very first Subject fixation, the IO fixation and the final Subject fixation were included in the analyses.

We analyzed the time-course of new fixations using two complementary methods. First, we estimated the probability of a new fixation towards the IO vs. distractor relative to the subject within discrete time bins of 150 ms windows, using a hierarchical, multilevel, logistic mixed-effects model (Gelman et al., 2012). This model was specified to estimate an overall mean log-odds of a new fixation for each 150 ms time bin, a global estimate of C-COMMAND and its variance for each time bin. To guard against multiple comparisons due to testing the C-COMMAND effect across several time bins, the corresponding contrasts were constrained through partial pooling (see Appendix A).

Second, as a descriptive counterpart to our primary, multilevel model, we modeled the time-course of differences in new fixations towards the IO vs. distractor (relative to the subject) using Generalized Additive Mixed-effects Models (GAMMs; Wood, 2017). GAMMs use splines which are smooth functions constructed from weighted and summed basis functions and used to map flexible, continuous curves over time. This continuous modeling was implemented to generate visualizations of the difference between the two conditions and to evaluate whether the difference exhibited a (non-)linear trend over time. Specifically, we fitted a difference spline model (Sóskuthy, 2017) over fine-grained 25 ms time bins for high temporal resolution using the `brms` package, which interfaces with the `mgcv` package in R (Wood, 2017).

The coding schemes and prior specifications were the same across all models, with the exception of the fixed effect of C-COMMAND. C-COMMAND was sum-contrast coded in the multilevel model (IO: 0.5, Distractor: -0.5) and treatment-coded in the difference spline model, with the IO taken as the reference level (IO: 0, Distractor: 1). All models included a maximal structure with random intercepts and slopes/splines for participants and items. Priors were chosen as weakly informative: Intercepts had a Gaussian prior with mean 0 and standard deviation of 1; the fixed effect had a Gaussian prior with a mean of 0 and standard deviation of 0.5; the standard deviation for the random effects had a half-Gaussian prior with a mean of 0 and a standard deviation of 1; and the correlation matrices in the multilevel model had a LKJ prior with a shape parameter of 2. In the spline model, time was center-scaled. We ran each model with four chains, each containing 6000 samples, 2000 of which were warm-up.

Lastly, we conducted formal hypothesis testing on the multilevel model using Bayes factors. As in divergence point analyses, we performed a prior sensitivity analysis evaluating four different prior sets. These sets crossed

two different a priori assumptions for the global effect of C-COMMAND (μ) with two assumptions for its standard deviation across time bins (σ_δ), as summarized in Table 2. The priors for the global population-level effect were zero-centered normal distributions with standard deviations of 0.3 and 0.5, corresponding to small and large magnitudes of the C-COMMAND effect, respectively. The priors for its standard deviation across time bins were half-normal distributions (truncated at zero) with standard deviations of 0.5 and 1.0, representing strong and weak shrinkage of the bin-specific slopes towards the global average, respectively.

Parameter	Prior width	Prior distribution	Assumed range (probability %)
Intercepts	–	Normal(0, 1)	[12.3, 87.7]
C-COMMAND (μ)	Small effect	Normal(0, 0.3)	[−14.3, 14.3]
	Large effect	Normal(0, 0.5)	[−22.7, 22.7]
SD of C-COMMAND (σ_δ)	Strong shrinkage	Half-normal(0, 0.5)	[50.0, 72.7]
	Weak shrinkage	Half-normal(0, 1)	[50.0, 87.7]
All other SDs	–	Half-normal(0, 1)	
Correlations	–	LKJ(2)	

Table 2: Priors used for the Bayes factor sensitivity analyses on the multilevel logistic mixed-effects model of new fixations. Four models were run crossing the two prior widths of the overall C-COMMAND effect (μ) and the two prior widths for its time-bin variability (σ_δ). Assumed ranges represent 95% prior probability mass on the probability scale (based on a 50% baseline new fixation proportion).

Below we present the results from the pooled dataset combining two experiments from the original study. Separate analyses conducted on individual source datasets yielded patterns qualitatively similar to those reported here (see Appendix B and Appendix C).

3. Results

Participants’ mean accuracy across filler items was 94.7%, suggesting that they remained attentive in the task. Trials with a track loss of 50% or more were removed from the analyses, which resulted in the loss of 0.008% of experimental items.

3.1. Response data

We first examined participants’ end-of-trial response data to reveal their ultimate interpretations for the anaphor. Table 3 shows the percentages of responses for each referent on the visual display. We found that while the

subject was chosen reliably more often in the Distractor than the IO condition ($\hat{\beta} = -1.26$ $[-1.70, -0.85]$, $\Pr(\hat{\beta} > 0) = 0$), IO was chosen reliably more often than the distractor ($\hat{\beta} = 1.31$ $[0.89, 1.76]$, $\Pr(\hat{\beta} > 0) = 1$). This confirms that subjects and IOs were interpreted as the anaphor’s antecedent significantly more often than the distractor.

	Subject %	IO/Distractor %	Unmentioned Plural % Singular %	
IO	80.2 (1.7)	18.8 (1.7)	0.6 (0.2)	0.4 (0.2)
Distractor	92.5 (1)	6.8 (0.9)	0.4 (0.2)	0.4 (0.2)

Table 3: Mean percentages (and SEs over by-participant means) for end-of-sentence anaphor interpretations for each referent.

All subsequent analyses were restricted to trials in which participants selected the subject as the antecedent of the anaphor. While the anaphor’s interpretation was ambiguous between the subject and the IO in the IO condition, the subject was the only available antecedent in the Distractor condition. By restricting our analyses to subject-response trials, we aimed to hold the final behavioral response constant in comparing looks towards a competing, grammatical IO antecedent against an unavailable distractor. This exclusion led to the removal of 13.7% of trials (302 out of 2,206).

3.2. Divergence point analyses

Figure 3 shows the mean gaze proportions towards the subject, IO/distractor and unmentioned plural referent, their estimated onsets of divergence points and the distributions of between-condition differences.

For the divergence from the subject, the divergence onset estimates showed a difference between the IO and Distractor conditions. The mean onset of divergence point was estimated at 469 ms [204, 735] for the IO condition, and 117 ms [5, 230] for the Distractor condition. The posterior mean for the between-condition difference was 352 ms [63, 641] ($\Pr(\hat{\beta} > 0) = 0.99$). The Bayes factor analyses showed moderate evidence for a between-condition difference across all tested priors (see Table 1 above) (BF_{10} range: 4.74–6.03). This indicates that the divergence from the subject was earlier in the Distractor relative to the IO condition.

For the divergence from unmentioned plural, we again observed a difference between conditions, but in the opposite direction. The onset of divergence was estimated at 192 ms [80, 305] for the IO condition, and 360 ms

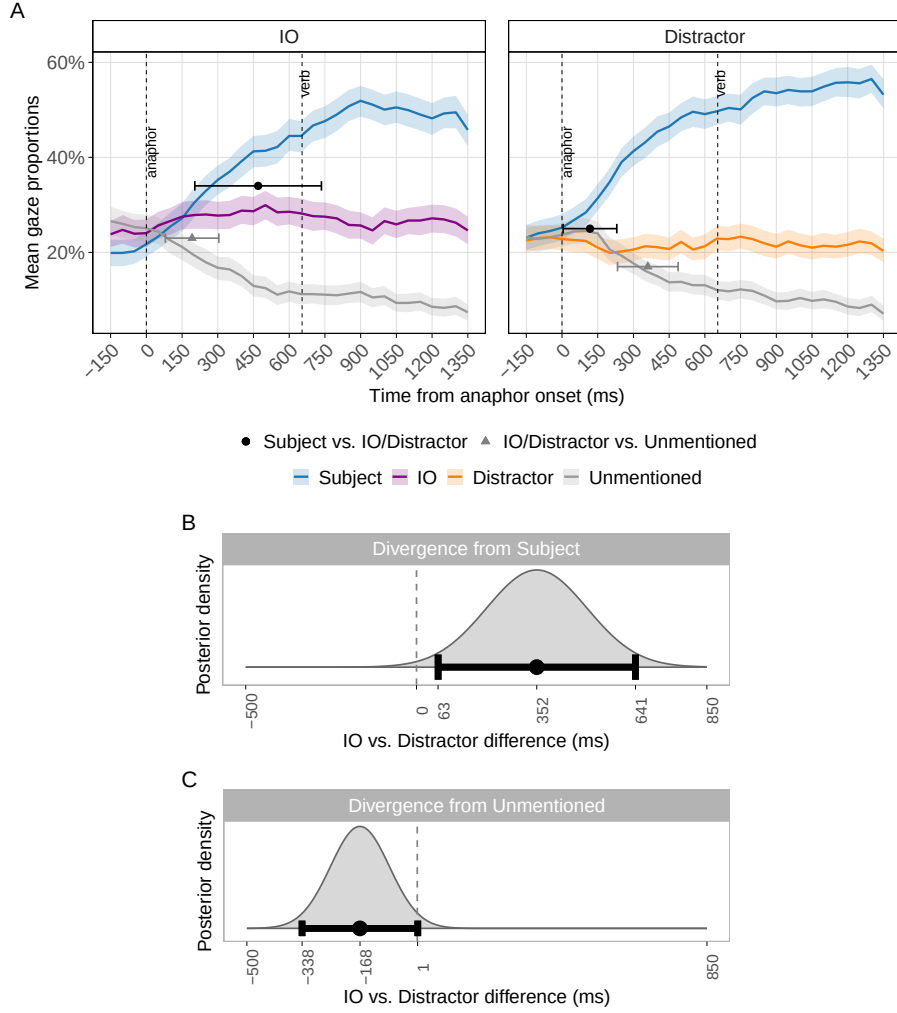


Figure 3: **A.** Gaze proportions for the critical referents, with estimated onsets of divergence points from the subject and unmentioned (plural) in the IO/Distractor conditions. Fixation curves show mean gaze proportions, and error bands show bootstrapped 95% CrIs. Proportions are time-locked to the beginning of the anaphor, extending to 1500 ms. Vertical dashed lines mark the 150 ms onsets beginnings of the anaphor and (embedded) verb windows. Point estimates indicate mean of divergence onsets in posterior distributions, with error bars showing their 95% CrIs. **B & C.** Posteriors of the between-condition differences in divergence from the subject and unmentioned referent, respectively. The point estimates show mean differences, with error bars showing their 95% CrIs. Vertical dashed lines mark a difference of 0.

[233, 488] for the Distractor condition. The posterior mean for the between-condition difference was -168 ms $[-338, 1]$ ($\Pr(\hat{\beta} > 0) = 0.974$). The Bayes factor analyses showed anecdotal evidence for a between-condition difference across all tested priors (BF_{10} range: 1.18–1.89). These results suggest that divergence from the baseline, unmentioned plural referent was delayed in the Distractor relative to the IO condition.

3.2.1. Discussion

Our divergence point analyses provide evidence that the target subject and target IO are reactivated more rapidly than the distractor. In comparing the time course of looks to IO and distractor against the subject, we see that subjects are distinguished from distractors as early as 117 ms following the anaphor onset. Conversely, in the IO condition, subject and IO receive an equal amount of fixations until approximately 469 ms, at which point looks to the subject start to increase significantly relative to the IO. The resulting 352 ms difference between these two conditions suggests that target subjects and IOs sustain a higher level of activation relative to distractors during the initial stages of anaphor resolution.

Turning now to the time-course of looks to the target IO and the distractor relative to the unmentioned, baseline referent, we see that the IO begins to diverge from baseline at approximately 192 ms, whereas the distractor does not show a reliable divergence until around 360 ms. Although the distractor curve appears descriptively flat in Figure 3, it reliably separates from the baseline as fixations to the baseline referent decline over time; we interpret this sustained elevation above the declining baseline as evidence of distractor reactivation. The posterior distribution indicates that the resulting 168 ms difference between the conditions is reliable ($\Pr(\hat{\beta} > 0) = 0.974$); thus we interpret our findings as evidence for a delay in reactivating the distractor relative to the IO. While the Bayes factor analyses provide only weak evidence for this difference, this is likely due to low statistical power.

It is worth bearing in mind that several caveats remain. First, gaze proportion analyses can be confounded by eye-movement behaviors carried over from earlier time windows, and as such they do not definitively inform us about the precise time-course of differences resulting from experimental manipulations. Second, while divergence point analyses indicate when looks to two referents begin to diverge, they fail to account for how baseline biases towards those referents might affect these estimates. A visual inspection of Figure 3 suggests a slight baseline bias towards the IO over the subject at

time 0, which is reversed in favor of the subject in the Distractor condition. Notably, in the comparisons against the unmentioned referent, the initial biases for the IO and distractor at time 0 seem smaller and go in the same direction across conditions. Nevertheless, although the baseline differences in our case seem slight, divergence point analyses as applied here do not take them into account. To address these limitations, we turn to analyses of new fixations, which better isolate eye-movement behaviors within discrete, smaller time windows.

3.3. New fixation analyses

Figure 4 shows new fixation counts on the subject, IO and distractor, starting from the onset of the anaphor extending to 6600ms. This time window includes the vast majority of the trials with a subject response, i.e., 85% in total. At a descriptive level, the overall frequency of new fixations decreased as time progresses and participants converged on a response, as is typical in the visual world paradigm. While new fixations on the subject remained consistently higher than those on the distractor, those on the subject and IO followed each other more closely. This was particularly evident in the early time windows following the anaphor onset, with the most pronounced difference appearing in the 150–300 ms time bin. Overall, participants were slower to select the subject as the antecedent of the anaphor in the IO relative to the Distractor condition ($\hat{\beta} = 0.19$ [0.13, 0.24], $\Pr(\hat{\beta} > 0) = 1$)⁴.

In the analyses we estimated the probabilities of new fixations on the IO/distractor vs. subject within the first 3 seconds after the anaphor onset because this time window covers the entire sentence plus at least an additional 1 second. Moreover, the significant drop in new fixations after 3 seconds leads to sparse data that may not be reliably modeled. Figure 5 illustrates the estimated probabilities of new fixations on the IO/distractor vs. subject, along with the estimated difference between the two conditions. The results from the multilevel and the difference spline model are in Tables 4 and 5, respectively.

The primary focus of the multilevel model⁵ was the global effect of C-

⁴Although participants could select a response after the anaphor onset, the experimental set-up required them to view the display while listening until the end of the sentence.

⁵is.IO/Distractor.fixated $\sim 0 + \text{TimeBin} + \text{C-COMMAND} + (0 + \text{C-COMMAND} | \text{TimeBin}) + (0 + \text{TimeBin} + \text{TimeBin:C-COMMAND} | \text{Subject}) + (0 + \text{TimeBin} + \text{TimeBin:C-COMMAND} | \text{Item})$

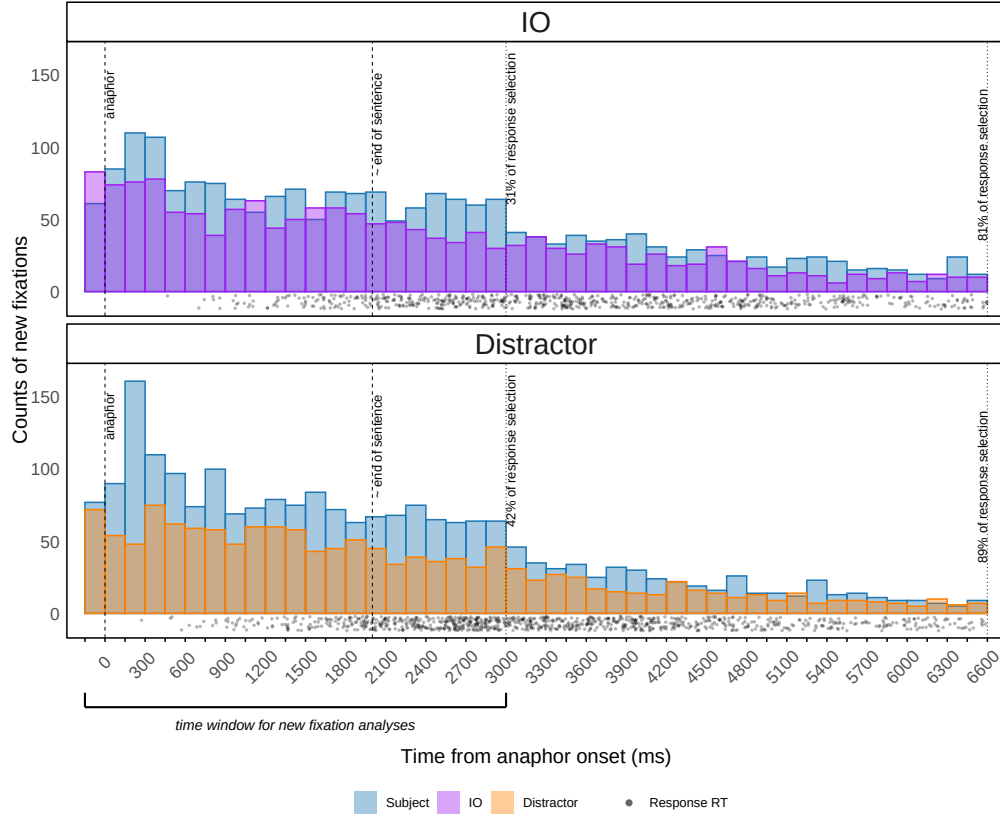


Figure 4: Counts of new fixations towards the subject, IO and distractor within 150 ms time bins time-locked to the anaphor onset (after the 150 ms shifting) and extending to 6600 ms. Gray dots below the x axis show individual response times. Vertical lines mark the end of the sentence (~ 2 s), proportions of trials with a response by 3 s and the time points by which 80% of the trials had a response. The bracket on the x axis shows the time window for the new fixation analyses.

COMMAND and its time-bin-specific deviations. While we report the intercept estimates, i.e., the grand mean log-odds of a new fixation towards the IO or distractor for each time bin, we do not interpret them directly as they do not test any hypothesis of interest.

Results from the multilevel model showed evidence for an overall effect of C-COMMAND, with more new fixations towards the IO relative to the distractor. This effect varied across the trial, with time-bin-specific slopes showing reliable effects in certain windows only. Following the anaphor onset, within

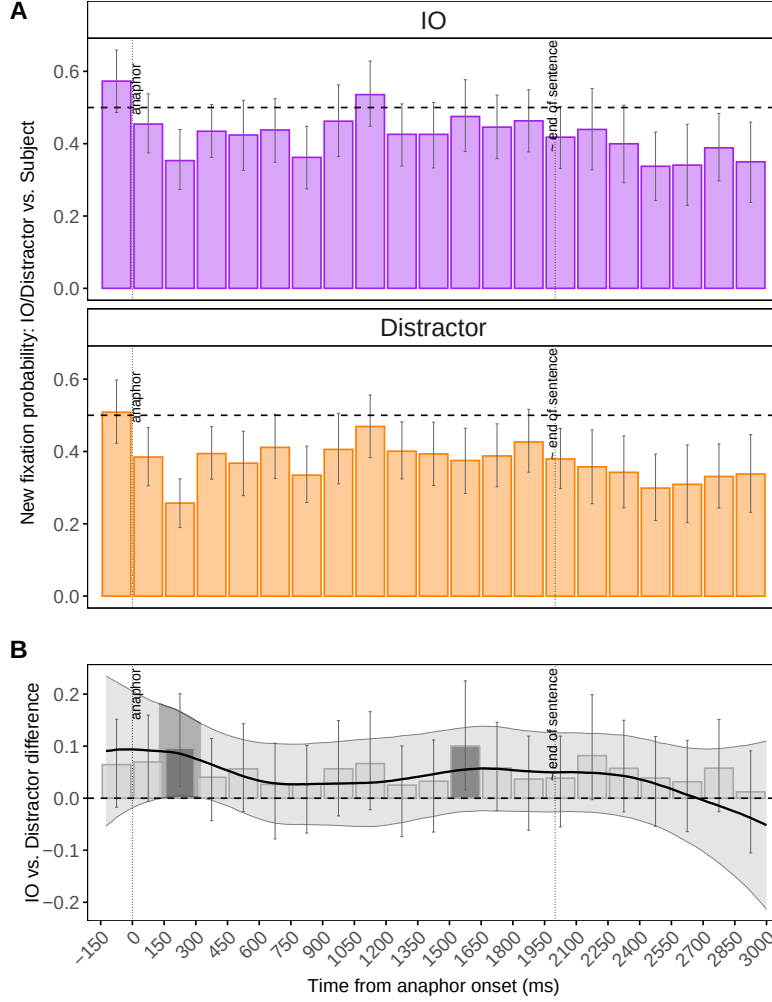


Figure 5: Time course of the estimated new fixation proportions on the IO/distractor vs. subject (**A**) and their difference (**B**), backtransformed to probability scale. Estimates are time-locked to the anaphor onset and extend to 3000 ms. In both panels bars show mean proportions and 95% CrIs extracted from the multilevel model on 150 ms time bins. Overlaid solid lines and shaded ribbons in (**B**) indicate the continuous posterior means and 95% CrIs extracted from the difference spline model. Vertical dotted lines mark the 150 ms shifted anaphor onset (0 ms) and the shifted end of experimental sentences (2000 ms). In (**A**), the horizontal dashed line at 0.5 marks equal probability of a new fixation on the IO/distractor vs. the subject. In (**B**), the horizontal dashed line at 0 marks no difference between the IO and Distractor conditions. Darker shaded bins and ribbons in (**B**) show a reliable difference between conditions, as indicated by $\Pr(\hat{\beta} > 0) \geq 0.975$.

Population-level coefficients					$\hat{\beta}$	SE	95% CrI	Pr($\hat{\beta}>0$)
C-COMMAND (μ)					0.22	0.08	[0.05, 0.38]	0.99
SD of C-COMMAND (σ_{δ})					0.20	0.11	[0.01, 0.42]	—
TimeBin-specific coefficients								
Intercepts	$\hat{\beta}$	SE	95% CrI	TimeBin-specific slopes	$\hat{\beta}$	SE	95% CrI	Pr($\hat{\beta}>0$)
−150–0ms	0.17	0.16	[−0.14, 0.49]	−150–0ms: C-COMMAND	0.26	0.17	[−0.07, 0.62]	0.95
0–150ms	−0.33	0.15	[−0.62, −0.03]	0–150ms: C-COMMAND	0.29	0.17	[−0.03, 0.67]	0.96
150–300ms	−0.84	0.15	[−1.14, −0.56]	150–300ms: C-COMMAND	0.46	0.23	[0.11, 0.96]	0.99
300–450ms	−0.35	0.13	[−0.61, −0.09]	300–450ms: C-COMMAND	0.17	0.16	[−0.18, 0.48]	0.86
450–600ms	−0.43	0.18	[−0.81, −0.09]	450–600ms: C-COMMAND	0.24	0.17	[−0.11, 0.60]	0.92
600–750ms	−0.31	0.16	[−0.62, 0.00]	600–750ms: C-COMMAND	0.11	0.19	[−0.32, 0.44]	0.74
750–900ms	−0.63	0.16	[−0.96, −0.33]	750–900ms: C-COMMAND	0.12	0.19	[−0.30, 0.45]	0.77
900–1050ms	−0.27	0.19	[−0.64, 0.09]	900–1050ms: C-COMMAND	0.23	0.18	[−0.14, 0.62]	0.91
1050–1200ms	0.01	0.16	[−0.29, 0.33]	1050–1200ms: C-COMMAND	0.27	0.19	[−0.09, 0.67]	0.94
1200–1350ms	−0.35	0.15	[−0.65, −0.07]	1200–1350ms: C-COMMAND	0.10	0.18	[−0.31, 0.42]	0.74
1350–1500ms	−0.37	0.17	[−0.71, −0.06]	1350–1500ms: C-COMMAND	0.14	0.18	[−0.27, 0.47]	0.79
1500–1650ms	−0.31	0.17	[−0.64, 0.02]	1500–1650ms: C-COMMAND	0.42	0.23	[0.07, 0.94]	0.99
1650–1800ms	−0.34	0.16	[−0.67, −0.02]	1650–1800ms: C-COMMAND	0.24	0.17	[−0.10, 0.61]	0.93
1800–1950ms	−0.22	0.16	[−0.53, 0.08]	1800–1950ms: C-COMMAND	0.15	0.18	[−0.25, 0.49]	0.82
1950–2100ms	−0.41	0.16	[−0.73, −0.12]	1950–2100ms: C-COMMAND	0.16	0.18	[−0.23, 0.50]	0.83
2100–2250ms	−0.42	0.21	[−0.85, −0.03]	2100–2250ms: C-COMMAND	0.35	0.22	[−0.01, 0.84]	0.97
2250–2400ms	−0.54	0.21	[−0.97, −0.14]	2250–2400ms: C-COMMAND	0.25	0.19	[−0.12, 0.65]	0.92
2400–2550ms	−0.77	0.20	[−1.19, −0.39]	2400–2550ms: C-COMMAND	0.18	0.20	[−0.25, 0.56]	0.85
2550–2700ms	−0.74	0.24	[−1.25, −0.30]	2550–2700ms: C-COMMAND	0.15	0.20	[−0.30, 0.51]	0.79
2700–2850ms	−0.58	0.18	[−0.96, −0.23]	2700–2850ms: C-COMMAND	0.25	0.19	[−0.11, 0.67]	0.92
2850–3000ms	−0.66	0.22	[−1.13, −0.25]	2850–3000ms: C-COMMAND	0.05	0.23	[−0.48, 0.40]	0.65

Table 4: Results of the Bayesian hierarchical, multilevel, logistic mixed-effects model on new fixations within discrete 150 ms time bins. $\hat{\beta}$ is the estimated mean of the posterior distribution over the parameter, given the data. Reliable effects of C-COMMAND, as indicated by $\text{Pr}(\hat{\beta}>0) \geq 0.975$, are in bold.

	$\hat{\beta}$	SE	95% CrI	Pr($\hat{\beta}>0$)
Intercept	−0.26	0.04	[−0.34, −0.18]	—
C-COMMAND: Distractor	−0.20	0.06	[−0.31, −0.09]	0.001
Time	−0.02	0.01	[−0.03, −0.01]	0.001
Time * C-COMMAND	0.01	0.01	[−0.0, 0.03]	0.96

Table 5: Results of the difference spline model on new fixations. $\hat{\beta}$ is the point estimate (mean) of the posterior distribution over the parameter, given the data. The reliable effect, as indicated by $\text{Pr}(\hat{\beta}<0) \geq 0.975$, is shown in bold.

the −150–150 ms time window, there was a trend for more new fixations towards the IO. This trend became statistically reliable within the 150–300 ms and 1500–1650 ms windows. A similar trend in the same direction was also observed in the 2100–2250 ms window, though this was not statistically reliable (see Appendix C for similar results on the source datasets).

The results of the Bayes factor analyses showed that across all tested priors on the global effect of C-COMMAND (see Table 2 above), there was moderate evidence for overall more new fixations towards the IO relative to the distractor ($\hat{\beta}$'s ≥ 0.21 , BF_{10} range: 3.55–5.31). Within the 150–300 ms and 1500–1650 ms time bins, there was evidence for more new fixations towards the IO across all models ($\text{Pr}(\hat{\beta}'\text{s} > 0) \geq 0.99$). There was no evidence for an effect in the other time bins ($\text{Pr}(\hat{\beta}'\text{s} > 0) \leq 0.97$).

Results from the difference spline model⁶ suggested evidence for an overall difference between conditions, with a higher probability of new fixations towards the IO relative to the distractor ($\text{Pr}(\hat{\beta} > 0) = 0.01$, $\text{BF}_{10} = 226.6$). There was also a trend for a non-linear shape of the difference smooth, but this trend was not statistically reliable ($\text{Pr}(\hat{\beta} > 0) = 0.96$, $\text{BF}_{10} = 0.08$). Extracted posterior distributions showed evidence for a higher probability of new fixations on the IO within the 125–325 ms window ($\hat{\beta}$'s ≥ 0.31 , $\text{Pr}(\hat{\beta}'\text{s} > 0) \geq 0.975$). While the extracted posteriors for the 1500–2150 ms time window showed a similar trend, they were not reliable ($\text{Pr}(\hat{\beta} > 0) \leq 0.92$).

3.3.1. Discussion

Our new fixation analyses provide further evidence that hierarchical structure shapes the time-course of access to potential antecedents in memory. Overall, the analyses on new fixations towards the IO/distractor show that the IO is accessed earlier and more frequently than the distractor when both are compared against the grammatically available subject. This prioritized access is evidenced by a higher probability of new fixations on the IO starting immediately after anaphor onset. While the continuous difference spline model descriptively characterizes the early trajectory of this distinction within the 125–325 ms window, the multilevel model evaluates the statistical evidence, reliably isolating this effect within the 150–300 ms bin. These results indicate that the c-commanding IO is accessed rapidly upon encountering the anaphor, whereas the distractor remains significantly less available than the subject during initial processing.

Beyond the initial stages of memory access, the results demonstrate that while structural information continues to modulate competition between hierarchically available referents during later stages of processing. This is evi-

⁶is.IO/Distractor.fixated $\sim$ C-COMMAND + s(Time, bs = “cr”, k = 30) + s(Time, by = C-COMMAND, bs = “cr”, k = 30) + s(Time, Subject, by = C-COMMAND, bs = “fs”, xt = “cr”, k = 8, m = 1) + s(Time, Item, by = C-COMMAND, bs = “fs”, xt = “cr”, k = 8, m = 1)

denced in the multilevel model by transient increases in new fixations on the IO relative to the distractor within the 1500–1650 ms and 2100–2250 ms windows, with more reliable evidence for the former. The 1500–1650 ms window roughly corresponds to the onset of the main verb at the end of the sentence, suggesting that the integration of the final verb may trigger a reevaluation of the interpretation or a period of sustained competition between two grammatically available referents. This localized effect is not reliable in the difference spline model, likely because the global smoothing function of the spline lacks the sensitivity required to capture short-lived spikes embedded within otherwise stable windows. Still, the spline model is descriptively informative, as it suggests (limited) evidence for a non-linear trend in the difference smooth across the entire 3-second time window, demonstrating that the IO-distractor difference dynamically changes over time.

4. General discussion

In this work we examined two alternative hypotheses regarding how hierarchical structural information is leveraged within working memory during language comprehension. Under the first one, structural information is used to selectively maintain hierarchically prominent, potentially c-commanding target items in a privileged, focal state for faster access; under the second, structural information is deployed at retrieval to gradually distinguish targets from structurally unavailable distractors. To adjudicate between these two, we used a published, visual world dataset to compare the time-course of reactivation for target antecedents against clause-mate distractors during the processing of the Turkish local reciprocal.

Our Bayesian analyses of gaze proportions revealed that target subjects and indirect objects were rapidly distinguished from distractors, as evidenced by their greater reactivation within the anaphor window. Importantly, although distractors exhibited some degree of transient reactivation, this was delayed relative to that of the indirect object by approximately 170 ms. Furthermore, new fixation analyses demonstrated that the proportion of new fixations directed towards the indirect object (relative to the subject) was reliably higher than that towards the distractor, with the most notable difference emerging as early as 150 ms after the anaphor onset. Taken together, these findings provide evidence for faster memory access to target antecedents over distractors.

Distinct speeds of memory access are associated with distinct representational states in memory (e.g., McElree et al., 2003; McElree, 2006; Oberauer, 2002). Following this line of reasoning, we interpret the faster access to structurally defined target antecedents over distractors as evidence for a functional divide in linguistic memory that is shaped by hierarchical syntactic information. Within this architecture, representations of linguistic items in structurally prominent positions, such as the local subjects and indirect objects in our stimuli, are actively maintained in a privileged, focal state that renders them immediately available for direct access (McElree et al., 2003; McElree, 2006). Conversely, distractors in structurally less prominent positions lack this privileged state and must be accessed via a slower, explicit, activation-based retrieval mechanism (Lewis and Vasishth, 2005; Lewis et al., 2006). Overall, these findings support a hybrid model of retrieval in which both direct-access and activation-based retrieval mechanisms operate in tandem during real-time dependency formation (see also Vasishth et al., 2019).

Broadly, our findings add to previous observations that various types of linguistic information, including syntactic, semantic or morphological, can modulate the time-course of memory access (Franck and Wagers, 2020; Johns et al., 2015; McElree, 2000; McElree et al., 2003; Ness and Meltzer-Asscher, 2019; Wagers and McElree, 2009; Wagers and Phillips, 2014; Wagers and McElree, 2022). The current work expands this literature by demonstrating that hierarchical syntactic information within a single clause shapes the time-course of memory access.

Methodologically, our work highlights the utility of the visual world paradigm for the study of temporal dynamics of memory access during real-time formation of syntactic dependencies. While the SAT or response-signal paradigms have been the primary tools for investigating similar questions (McElree and Doshier, 1989; McElree et al., 2003; Wagers and McElree, 2022), evidence suggests that such techniques, as deployed in the sentence processing literature, may lack the statistical power that is necessary to detect fine-grained differences in processing speed (Logačev and Bozkurt, 2021; Pankratz et al., 2021). Other studies have used reaction-time measures from tasks such as self-paced reading or listening or sentence-picture matching to measure retrieval dynamics (Lissón et al., 2021, 2023; Nicenboim and Vasishth, 2018; Patil et al., 2016). Here we contribute to this methodological landscape by leveraging the visual world paradigm, which offers high temporal resolution and continuous sensitivity to underlying activation dynamics within a naturalistic auditory comprehension task.

4.1. Hierarchical structural information in focal attention

What is the memory mechanism that enables immediate, privileged access to hierarchically prominent, c-commanding linguistic items within a local clause? Recent structurally oriented sentence processing models posit that linguistic items are encoded and maintained through transient bindings with their syntactic positions, which are subsequently accessed by cues linked to these positions (Cho et al., 2017, 2020; Keshev et al., 2025; Logačev and Vasishth, 2011). Bridging these models with theories of focal attention, we propose that these item-position associations serve a broader organizational function, such that abstract syntactic positions of syntactically prominent items, such as the local subjects and indirect objects in our study, are dynamically bound to a structural template that is actively held in a privileged, focal memory state. It is this structural template that keeps hierarchically prominent representations concurrently available for further processing, without a need for an explicit retrieval process (e.g., McElree, 2006; Oberauer, 2002).

Such a structural template maintained within the focal memory state need not *necessarily* encode an exhaustive representation of linguistic items. Instead, it may primarily serve as a mapping that binds structurally prominent items to their syntactic coordinates, while potentially maintaining only a restricted subset of item-specific features. This mechanism can be formalized with the structured *region of direct access* in Oberauer’s (2002, 2009, 2019) working memory architecture. In this architecture, using dynamic bindings between content (e.g., lexical items) and context (e.g., structural positions defining the syntactic structure and hierarchical relations within a clause), the region of direct access builds and maintains spatially structured representations of a narrow set of activated items available for immediate access. A conceptually similar mechanism has been proposed by Wagers and McElree (2013), who hypothesize a *global focal attention* preserves a pruned representation of the global syntactic context, maintaining abstract, structural coordinates, alongside a subset of item-specific features. They further suggest that due to capacity limitations, the representations maintained within this privileged state are restricted to certain structural positions (such as those for the local, c-commanding arguments in our stimuli), along with some individual features.

Consequently, during the resolution of an anaphor-antecedent dependency like the one in the present study, this structural template can be immediately leveraged to rapidly link the anaphor with its local, c-commanding

antecedent, while the retrieval and integration of its remaining (e.g., full morphological or lexical) content follows this initial structural linking. Items in structurally less prominent positions, such as the deeply embedded distractors in our study, are by hypothesis excluded from this focal structural template and remain in a passive memory state. This passive state accommodates a larger set of representations, but at the expense of slower, explicit retrieval for reactivation of items contained therein.

In addition to explaining the difference in time course we observed in our analyses, this architecture also accommodates several existing findings in the literature. First, it offers an explanation for why the early stages of resolving a local anaphor are primarily guided by structural constraints, exhibiting only subtle and often inconsistent or delayed influence from feature-matching, unavailable distractors (e.g., Dillon et al., 2013; Kush and Phillips, 2014; Jäger et al., 2020; Phillips et al., 2011; Sturt, 2003, cf. Badecker and Straub, 2002; Parker and Phillips, 2017). In our account, because local, c-commanding target antecedents are actively maintained within a privileged focal state, they are accessed directly. This obviates the need for explicit retrieval operations that would otherwise induce similarity-based interference from distractors with matching features.

Second, this architecture provides a novel reinterpretation of the “local advantage” in anaphor processing, characterized by faster access to local over non-local antecedents (Arvindam, 2025; Dillon, 2011; Dillon et al., 2014, 2016). To account for this advantage, Dillon and colleagues proposed a “local search” mechanism that employs structural cues to prioritize local arguments during initial memory access (Wagers, 2008). In our architecture this speed advantage would emerge because local co-arguments occupy the focal state in working memory, rendering them available for immediate, fast access. This explanation is further supported by electrophysiological evidence showing additional retrieval costs for non-local dependencies (Dillon, 2011).

Third, our account captures a broader range of findings demonstrating that certain non-structural information, such as syntactic category, animacy or grammatical number, can be actively maintained within a privileged state and accessed rapidly during dependency formation (Franck and Wagers, 2020; Johns et al., 2015; McElree, 2000; McElree et al., 2003; Ness and Meltzer-Asscher, 2019; Wagers and McElree, 2009; Wagers and Phillips, 2014; Wagers and McElree, 2022) (see the Introduction). These patterns can be accommodated if we assume, following Wagers and McElree (2013), that the directly accessible focal state includes a restricted set of non-structural features

alongside a structured representation of prominent items. Which individual feature is maintained in this focal state appears to depend on a range of factors, including the type of the dependency being constructed and the local structural configurations involved, which we take up in the next section.

The proposed architecture also has broader implications for how working memory is deployed for structured memory access across a range of linguistic dependencies. Structured memory access is suggested as a defining property of dependencies that are syntactically constrained, such as the local anaphor examined here (Dillon, 2011; Dillon et al., 2013; Wagers, 2008). We propose that maintaining a structural template in focal attention is one potential strategy to mediate this structured access. If this is correct, then we should find evidence that the use of this mechanism extends to other structurally constrained dependencies. Subject-verb thematic dependencies are one such example, as they require interpreting a verb relative to its thematic subject in a certain structural position, as in *The editor of the author laughed*. Supporting our prediction, evidence suggests that thematic subjects are rapidly accessed via cues targeting hierarchical position or structural subjecthood (Arnett and Wagers, 2017; Kogan and Wagers, 2026; Mertzen et al., 2023; Schoknecht et al., 2025; Van Dyke, 2007; Van Dyke and McElree, 2011). Within our architecture, this is due to the active maintenance of the subject or its relevant features within the structural template in focal attention, which is deployed for rapid access at the point of the verb. This claim is supported by time-course studies demonstrating faster access to subjects relative to other nouns in a sentence (e.g., Franck and Wagers, 2020).

We note that focal attention may not be the primary informational matrix for processing all types of prominent information. To date, empirical work has not shown that discourse- or prosody-level prominence modulates which linguistic item is maintained within focal attention (e.g., Foraker, 2007; Foraker and McElree, 2007, cf. Kush et al., 2019). For example, Foraker and McElree (2007) demonstrated evidence for higher accuracy in interpreting the pronoun *he* when it referred back to a discourse-prominent antecedent in a clefted structure as in *‘It was the ardent boyfriend who contemplated the engagement ring. He stared’* relative to a non-prominent antecedent as in *‘What the ardent boyfriend contemplated was the engagement ring. He stared’*. Crucially, however, discourse prominence did not alter the speed of memory access for the antecedent. Similar patterns were found for prominence driven by spoken prosodic stress during listening (Foraker, 2007), where prosodic stress increased retrieval accuracy, but it did modulate retrieval

speed. Overall, these findings are consistent with our proposal that maintenance in focal attention is selectively constrained by certain types of linguistic prominence and features.

4.2. The scope of focal attention in sentence processing

Our finding that the structural coordinates of syntactically prominent items occupy a privileged memory state raises broader questions regarding how the functional architecture of working memory serves online sentence processing. Specifically, how is complex linguistic input dynamically partitioned into discrete representational states, and what determines which information sources are prioritized for immediate access? Furthermore, under what conditions do these structural or featural representations remain available over the course of processing?

Accumulating evidence suggests that the contents of focal attention do not map straightforwardly onto rigid structural boundaries such as clause nodes (cf. McElree et al., 2003) or phrasal projections (cf. Lewis and Vasishth, 2005). Rather, focal attention appears capable of selectively maintaining particular linguistic features of a global structure, independently of the structural units with which these features are associated (e.g., Wagers and McElree, 2013; Wagers and Phillips, 2014; Wagers and McElree, 2022). Current empirical evidence indicates that the dimensions of linguistic information held within this focal state span syntactic, semantic and morphological information (see the Introduction for a review). Taken together, the feature composition of focal attention appears to vary dynamically, allowing distinct features to be selectively prioritized depending on the dependency being resolved.

Once isolated, the maintenance of individual features in focal attention appears to depend on the structural relations they participate in as the sentence unfolds. For instance, local structural dependencies, such as an intervening relative clause or complex nominal modifiers, can compete for focal attention and displace previously encoded matrix subjects or DP-internal features from direct access (Johns et al., 2015; McElree et al., 2003; Wagers and McElree, 2022). Other features, by contrast, are more resilient. For example, syntactic category of a *wh*-filler remains readily available across multiple clausal boundaries during the processing of long-distance dependencies (Wagers and Phillips, 2014). The present study expands our understanding of the role of structural information in memory by demonstrating that positional

information associated with c-commanding antecedents can be actively kept in focal attention throughout the processing of a local clause.

A linguistic feature’s eligibility for focal attention may also be determined by its diagnostic value for upcoming structural relationships (Wagers and McElree, 2022). In prospective dependencies, such as filler-gap constructions or clefts, early-introduced elements hold high predictive value for anticipating downstream items, which may license their active maintenance in focal attention (McElree, 2000; Wagers and Phillips, 2014). By contrast, anaphor-antecedent dependencies examined here are retrospective because the appearance of a local argument does not generate a deterministic prediction that a matching anaphor will appear later in the sentence. Despite the lack of a prediction, our results demonstrate that hierarchical structural information can nevertheless be sustained within focal attention.

Abstract structural information of prominent items in a local clause may be prioritized not because it serves a single, specific dependency, but because it is broadly diagnostic for a wide range of upcoming linguistic relations. Given the capacity limits on focal attention, the parser may strategically prioritize global syntactic information over fine-grained, dependency-specific details, particularly in the absence of strong predictive features. This aligns well with a substantial body of evidence that sentence processing mechanisms rely primarily on syntactic information (e.g., Brennan et al., 2012; Clifton et al., 2003; Dillon, 2011; Frazier and Fodor, 1978; Kush, 2013; Phillips et al., 2011; Staub and Clifton Jr., 2006; Sturt, 2003; Wagers, 2008), with the current study offering a mechanistic account of how this syntactic primacy is mediated via a privileged, focal state in memory. To evaluate the generalizability of the claims made here, future research should investigate a wider range of constructions to determine the extent to which structural information is prioritized within memory representations and access, and how it interacts dynamically with other linguistic information.

4.3. Alternative accounts

Is it possible for a working memory architecture that does not invoke a focal/non-focal distinction to capture our data? While acknowledging that we cannot definitively rule this out, it remains unclear how a cue-based account, in which all items—targets and distractors alike—are stored uniformly and accessed via an activation-based retrieval mechanism (Lewis and Vasishth, 2005; Lewis et al., 2006), can independently provide a principled explanation for our findings. In these models, focal attention is restricted to a

single phrase-sized chunk (i.e., XP) currently being processed; all other items remain subject to explicit retrieval and are checked against retrieval cues in parallel. In a standard architecture where structural and non-structural (e.g., number) cues are weighted in an additive scheme, both targets and distractors should then exhibit a *simultaneous* increase in reactivation due to their (partial) feature match. However, this fails to account for our finding that the reactivation onset for the feature-matching, plural distractor is delayed by around 170 ms relative to the IO, when both are compared against the baseline referent.

It may be possible to capture this effect if we suppose that structurally informed cues serve a gating function via a multiplicative cue combination scheme, which would nullify the effect of non-structural cue matches (see Dillon et al., 2013; Kush et al., 2015; Parker and Phillips, 2017; Parker, 2019a; Van Dyke and McElree, 2011, for extended discussions). While this could explain the target’s advantage in initial memory access over distractors, it would fail to explain how the distractor could accrue any reactivation at a later time point.

To do this, one could posit a multi-stage retrieval model where an initial structural gating mechanism is followed by an additive scheme that attends to featural cues. However, it is unclear what would motivate such a shift in cue combinatorics, given that the distractors occupy structurally embedded and discourse-wise non-prominent positions in our configurations. Thus, while we consider a uniform, activation-based model unlikely to account for the full range of our findings, we acknowledge that this possibility warrants further exploration through computational simulations (see also Lissón et al., 2021, 2023; Nicenboim and Vasishth, 2018; Patil et al., 2016; Parker, 2019b).

5. Conclusion

In this work we examined the time-course of reactivation for structurally prominent target referents against distractors during the real-time resolution of a Turkish local anaphor, using a published, visual world dataset. Our analyses of gaze proportions and new fixation probabilities demonstrated rapid access to target referents during the initial stages of processing, whereas distractors exhibited weaker reactivation with a significant temporal delay. We interpret the temporal advantage of targets as evidence that structural information directly shapes the organization of linguistic representations in memory. We propose that a structured representation of syntactically promi-

nent linguistic items within a clause is preserved in a privileged, focal state in working memory, which allows key syntactic information to be kept immediately available for on-going processing. Items in less prominent (i.e., hierarchically embedded) positions are kept in a passive state, requiring re-activation via a slower retrieval mechanism. Overall, these findings advance our understanding of how syntactic structure guides real-time sentence processing, proposing the focal attentional state as a mediator for structured access.

Data Availability

The data and scripts for statistical analyses are available at the following link: https://zenodo.org/records/22234539?preview=1&token=eyJhbGciOiJIUzUxMiJ9.eyJpZCI6ImNiYjM5MTM0LWMzMdktNGFhNS1hOGI2LWUwODBiMTJlYjM5OCIsImRhdGEiOnt9LCJyYW5kb20iOiI3OGMzMdJlMmFhOWUwZjlkNzZkOGJmZWUyMmNlZTUyMCJ9.NE10r6DIGzUfNMgg_ibIxvi-TgwvxKPQ0tNeIhF9NuYCBsp-mMpo6lu1jDalic8dmFFShEzKpzutTbpCiPCKwQ

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Appendix A. Hierarchical multilevel model on new fixations

The equation in A.1 shows the formula of the hierarchical, multilevel, logistic mixed-effects model on new fixation probabilities:

$$\begin{aligned} \text{logit}(p_n) &= \gamma_{j[n]} + u_{subj[n],j[n],1} + w_{item[n],j[n],1} + \\ &\quad \text{C-COMMAND}_n \cdot (\delta_{j[n]} + u_{subj[n],j[n],2} + w_{item[n],j[n],2}) \\ \delta_j &\sim \text{Normal}(\mu, \sigma_\delta^2) \end{aligned} \quad (\text{A.1})$$

where p_n is the probability that observation n is a new fixation on the IO (or distractor), $j[n]$ indexes the 150-ms time bin, and $subj[n]$ and $item[n]$ index the participant and item associated with observation n , respectively. The term $\gamma_{j[n]}$ represents the baseline log-odds of a new fixation for time bin j , and C-COMMAND_n denotes the experimental condition of observation n . The coefficient $\delta_{j[n]}$ represents the C-COMMAND effect for time bin j , which is assigned a shared distribution centered at the global average effect μ , with variance σ_δ^2 determining the degree of partial pooling across time bins. Participant and item adjustments across time bins for intercepts and C-COMMAND effects are captured by $u_{subj[n],j[n]}$ and $w_{item[n],j[n]}$, respectively.

This hierarchical framework implements partial pooling for the C-COMMAND effect across time bins by shifting point estimates and their corresponding intervals closer to each other when necessary (e.g., in the case of noise or sparse data). Rather than treating each discrete 150 ms window as an isolated, independent effect, the model recognizes that all time bins measure the same underlying C-COMMAND effect. We prefer this multilevel modeling approach over a nested one (i.e, TimeBin:C-COMMAND) because via the regularization of the estimates under a shared variance parameter (σ_δ^2), the multilevel approach controls for multiple comparisons by mitigating Type S (sign) and Type M (magnitude) errors. Consequently, the model yields conservative estimates without requiring statistical power-reducing post-hoc adjustments (e.g., Bonferroni corrections) (see Gelman et al., 2012, for further details).

Appendix B. Divergence point analyses on the source experiments

We conducted divergence point analyses on the source datasets from Experiments 2 and 3 separately, following the steps in the main analyses. Table B.6 shows estimated divergence point onsets for the IO and distractor when compared against the subject and the baseline, unmentioned plural referent.

		Experiment 2		Experiment 3	
		Mean (ms)	95% CrI	Mean (ms)	95% CrI
Divergence from Subject	IO	809	[452, 1166]	403	[156, 649]
	Distractor	260	[24, 495]	119	[13, 225]
	Difference	549	[122, 977]	284	[15, 552]
Divergence from Unmentioned	IO	223	[66, 380]	285	[125, 446]
	Distractor	585	[137, 1033]	398	[218, 577]
	Difference	-362	[-838, 113]	-112	[-354, 129]

Table B.6: Summary of divergence onset estimates and between-condition differences for Experiments 2 and 3. The Mean (ms) column shows means of estimated divergence onsets for the IO and Distractor conditions and between-condition differences.

For divergence from the subject, both experiments replicated the qualitative pattern observed in the combined dataset. The difference between the two conditions suggested reliable evidence for a delayed divergence onset in the IO relative to the Distractor condition (Experiment 2: 549 ms [122, 977]; Experiment 3: 284 ms [15, 552]), which was supported by anecdotal to moderate evidence (Experiment 2: $\Pr(\hat{\beta} > 0) = 0.99$, $\text{BF}_{10} = 10.7$; Experiment 3: $\Pr(\hat{\beta} > 0) = 0.98$, $\text{BF}_{10} = 2.40$). Notably, the divergence onsets were overall earlier in Experiment 3 than in Experiment 2, which may reflect differences in within-experiment biases.

For divergence from the unmentioned referent, the individual datasets again mirrored the combined dataset. The difference between the two conditions showed a trend for an earlier divergence onset in the IO relative to the Distractor condition (Experiment 2: -362 ms [-838, 113]; Experiment 3: -112 ms [-354, 129]), though this evidence was at best anecdotal (Experiment 2: $\Pr(\hat{\beta} > 0) = 0.93$, $\text{BF}_{10} = 1.52$; Experiment 3: $\Pr(\hat{\beta} > 0) = 0.82$, $\text{BF}_{10} = 0.38$). While the statistical evidence in the individual experiments was weak, the direction of the effect is consistent and mirrors the analyses on the combined dataset.

Appendix C. New fixation analyses on the source experiments

We ran multilevel, logistic mixed-effects models on new fixation probabilities in Experiments 2 and 3 separately, following the steps in the main analyses. In line with the results reported for the combined dataset, there were overall more new fixations towards the IO relative the distractor, which showed variation across time bins (C-COMMAND μ : Experiment 2: $\hat{\beta} = 0.29$ [-0.01, 0.59], $\Pr(\hat{\beta} > 0) = 0.97$, $\text{BF}_{10} = 1.94$; Experiment 3: $\hat{\beta} = 0.19$ [0.01,

0.37], $\Pr(\hat{\beta} > 0) = 0.98$, $\text{BF}_{10} = 1.55$; C-COMMAND σ_δ : Experiment 2: $\hat{\beta} = 0.27$ [0.01, 0.66]; Experiment 3: $\hat{\beta} = 0.15$ [0.01, 0.39]).

The most reliable effect across both datasets was an increase in new fixations towards the IO relative to the distractor within the 150-300 ms time bin (Experiment 2: $\hat{\beta} = 0.58$ [0.05, 1.39], $\Pr(\hat{\beta} > 0) = 0.98$; Experiment 3: $\hat{\beta} = 0.27$ [-0.03, 0.71], $\Pr(\hat{\beta} > 0) = 0.96$). Results from Experiment 2 also yielded evidence for more new fixations towards the IO within the 1500-1650 ms time bin ($\hat{\beta} = 0.54$ [0.01, 1.41], $\Pr(\hat{\beta} > 0) = 0.97$). There were no other effects ($\hat{\beta}$'s ≤ 0.46 , $\Pr(\hat{\beta}'\text{'s} > 0) \leq 0.95$).